

```

Getting Started with PySpark.ipynb
File Edit View Insert Runtime Tools Help Cannot save changes

+ Code + Text Copy to Drive

38 packages can be upgraded. Run 'apt list --upgradable' to see them.
tar: spark-3.2.1-bin-hadoop3.2.tgz: Cannot open: No such file or directory
tar: Error is not recoverable: exiting now
Collecting pyspark
  Downloading pyspark-3.5.1.tar.gz (317.0 MB)
    317.0/317.0 MB 3.6 MB/s eta 0:00:00
  Preparing metadata (setup.py) ... done
Requirement already satisfied: py4j==0.10.9.7 in /usr/local/lib/python3.10/dist-packages (from pyspark) (0.10.9.7)
Building wheels for collected packages: pyspark
  Building wheel for pyspark (setup.py) ... done
  Created wheel for pyspark: filename=pyspark-3.5.1-py2.py3-none-any.whl size=317488493 sha256=0e2889c34dbdac112b6792bc74bc5183499b311be253915270dbf06a00250b4f
  Stored in directory: /root/.cache/pip/wheels/80/1d/60/2c256ed38ddc2fdd93be545214a63e02fbd8d74fb0b7f3a6
Successfully built pyspark
Installing collected packages: pyspark
Successfully installed pyspark-3.5.1
Requirement already satisfied: py4j in /usr/local/lib/python3.10/dist-packages (0.10.9.7)
SparkSession - in-memory
SparkContext
Spark UI
Version
v3.5.1
Master
local[*]
AppName
Our First Spark Example

```

Figure 3: Verifying the installation of PySpark

```

spark= SparkSession \
    .builder \
    .appName("Our First Spark
Example") \
    .getOrCreate()

spark

```

Step 5. Verify installation: Figure 3 shows the successful installation of PySpark.

Step 6. Create a test DataFrame (DF): DF is a fundamental data structure that represents a structured, immutable, and distributed collection of data. For example, you can create a PySpark DataFrame with the sample shown in Figure 4.

PySpark is a versatile and powerful tool for processing and analysing Big Data. Its distributed computing capabilities, combined with the ease of use and flexibility of the Python programming language, makes it an asset for data professionals and organisations dealing with large scale data processing tasks. In short, PySpark is a gateway to a world of possibilities where data-driven insights can transform businesses and drive success. **END**

Creating a Test Spark DataFrame

```

from pyspark.sql import SparkSession
from pyspark.sql import Row

# Create a Spark session
spark = SparkSession.builder \
    .appName("TestApp") \
    .getOrCreate()

# Create a list containing the string
data = [("OSFY is Asia's leading IT publication. Launched in February 2003 (as Linux For You)",)]

# Create an RDD from the list
rdd = spark.sparkContext.parallelize(data)

# Create a DataFrame with one column named 'DataFrame_Output'
df = rdd.map(lambda x: Row(text=x[0])).toDF()

# Show the DataFrame
df.show(truncate=False)

# Stop the Spark session
spark.stop()

```

```

+-----+
|text|
+-----+
|OSFY is Asia's leading IT publication. Launched in February 2003 (as Linux For You)|
+-----+

```

Figure 4: Creating a test PySpark DataFrame

References

- <https://spark.apache.org/downloads.html>
- <https://spark.apache.org/>
- <https://spark.apache.org/docs/latest/>

By: Dr Aditya Bhardwaj

The author is a B. Tech, M. Tech, and PhD in CSE. He works as assistant professor in the School of Computer Science Engineering and Technology at Bennett University, Greater Noida. He is experienced in cloud computing and open source technology.